

Assignment 03 – Dynamic Dropdowns & Checkboxes

1. Module Name

Module 1 – Automation with Selenium

2. Experiment Title

Dynamic Dropdowns & Checkboxes

3. Problem Statement

Interact with multiple checkboxes and a dynamic autocomplete dropdown. Select and verify the checkboxes, enter partial text into the autocomplete field, iterate through the suggestions, and select the matching country.

4. Objective

To automate checkbox controls and dynamic autocomplete dropdowns using Selenium and verify the selected values.

5. Tools, Software and Concepts Used

- Python 3.x
- Selenium WebDriver
- Google Chrome
- Visual Studio Code
- By.ID
- By.CSS_SELECTOR
- WebDriverWait
- expected_conditions
- is_selected()
- Loops
- assert statement

6. Source Code / Implementation Steps

1. Open the Automation Practice page.
2. Select Checkbox Option 1 and Option 2 if required.
3. Verify both checkboxes using is_selected().
4. Enter ind in the autocomplete field.
5. Wait for the dynamic suggestions.
6. Loop through the suggestions and select India.
7. Verify that India is selected.
8. Print the successful execution result and close the browser.

Source Code

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC

driver = webdriver.Chrome()

try:
    driver.get("https://rahulshettyacademy.com/AutomationPractice/")
    driver.maximize_window()

    print("Automation Practice page opened.")

    checkbox1 = driver.find_element(By.ID, "checkBoxOption1")
    if not checkbox1.is_selected():
        checkbox1.click()

    checkbox2 = driver.find_element(By.ID, "checkBoxOption2")
    if not checkbox2.is_selected():
        checkbox2.click()

    assert checkbox1.is_selected()
    assert checkbox2.is_selected()

    print("Checkbox Option 1 selected:", checkbox1.is_selected())
    print("Checkbox Option 2 selected:", checkbox2.is_selected())

    autocomplete = driver.find_element(By.ID, "autocomplete")
    autocomplete.send_keys("ind")

    wait = WebDriverWait(driver, 10)
    suggestions = wait.until(
        EC.visibility_of_all_elements_located(
            (By.CSS_SELECTOR, ".ui-menu-item div")
        )
    )

    country_selected = False

    for suggestion in suggestions:
        if suggestion.text.strip().lower() == "india":
            suggestion.click()
            country_selected = True
            break

    assert country_selected
    assert autocomplete.get_attribute("value") == "India"

    print("Selected country:", autocomplete.get_attribute("value"))
    print("Autocomplete selection successful.")
    print("Assignment 03 completed successfully.")

    input("Press ENTER to close the browser...")

finally:
    driver.quit()
```

7. Output / Labelled Screenshots

- Screenshot 1 – Initial Automation Practice page
- Screenshot 2 – Checkbox Option 1 and Option 2 selected
- Screenshot 3 – India selected from the autocomplete dropdown
- Screenshot 4 – VS Code terminal showing successful execution

Screenshots are stored in the Screenshots folder for this assignment.

8. Result

The required checkboxes were selected and verified successfully. The dynamic autocomplete suggestions were handled and India was selected and verified.

9. Observation

Checkboxes can be verified using `is_selected()`. Dynamic autocomplete controls require the script to wait for suggestions and inspect the generated list before selecting the required option.

10. Conclusion

The experiment successfully demonstrated Selenium automation of checkbox controls and dynamic autocomplete dropdowns using locators, explicit waits, loops and assertions.